PE and Private Credit Cross-Dealing

Do Relationship Lenders Price Differently?
Evidence from Synthetic Deal Data

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1. Research Question and Setting

Large alternative asset managers increasingly operate both private equity (PE) and private credit (PC) arms within the same umbrella organisation. This creates a setting in which firms can simultaneously own companies and lend to those companies – or to each other's portfolio companies. This memo investigates whether debt pricing (measured by loan spread in basis points) differs systematically based on the relationship between the PE sponsor and the lending arm at the time of the loan. The central question is: after controlling for deal year, does a repeated, affiliated, or reciprocal relationship between PE and PC yield a different loan spread than an arm's-length transaction with an unrelated lender? A companion question asks whether companies that borrow from relationship lenders exhibit different entry characteristics or exit outcomes. The analysis uses two synthetic datasets. **Atlas** provides 315,238 deal-level records covering buyouts, loans, IPOs, and exits across 112,276 companies. **Summit** provides 48,317 fund-investment records from general partners, including entry financials and eventual returns. The two datasets are linked by company name after standardisation.

2. Data and Panel Construction

2.1 Funnel from Raw Events to Analytical Sample

Constructing the analytical sample required a multi-step transformation. Atlas records deal events without explicit ownership panels, so ownership spells must be inferred from event sequences.

2.2 Ownership Spell Construction

A PE ownership spell begins when a firm executes a Buyout/LBO or Secondary Transaction - Private deal and ends at the earliest of: a subsequent buyout of the same company (secondary buyout), an IPO, a merger or acquisition, a corporate asset purchase, bankruptcy (Chapter 7 or 11), or an out-of-business event. Deals tagged *Secondary Buyout* in the `DealType2` field perform double duty - they close the prior owner's spell and open the new owner's spell on the

Table 1: Funnel from Atlas Events to Cross-Deals

Stage	**N**	**% of prior**
Atlas rows (raw)	315,238	
Buyout / spell-start events	43,517	13.8%
PE ownership spells constructed	38,204	87.8%
Debt deal events (raw)	31,349	
PE-backed debt deals (date-matched)	18,671	59.6%
Cross-umbrella deals	14,228	76.2%
Same-umbrella affiliate deals	4,443	23.8%

Note: Spell-start events include Buyout/LBO and Secondary Transaction - Private (DealClass = Private Equity). Growth-equity deals excluded.

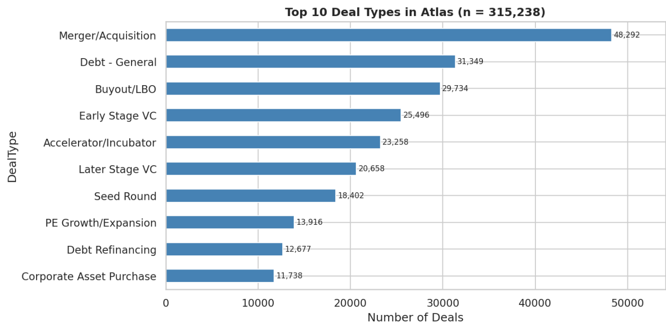

Figure 1: Top 10 Deal Types in Atlas ($n = 315{,}238$)

same date. Growth-equity investments (PE Growth/Expansion) are excluded from spell starts by design, because they typically represent minority stakes without full control. This is a configurable parameter in the pipeline. Spells without an observed exit event remain open; subsequent loans are attributed to the active PE sponsor. Raw PE sponsor and credit provider names are cleaned by lowercasing, stripping punctuation, and removing common legal suffixes (LLC, LP, Group, Partners, Capital, Management, etc.). The umbrella label is the first two meaningful tokens of the cleaned name. Four relationship types are defined:

- **Same-umbrella affiliate:** the PE sponsor and the credit provider share the same umbrella label.
- **Prior-reciprocal:** the lender has lent to the sponsor's portfolio companies *before* this deal date, and the sponsor has lent to the lender's portfolio companies in prior history. Constructed using only pre-deal information, this is the preferred causal measure.
- **Reciprocal (future only):** reciprocal dealing observed in the dataset but not yet at the time of this deal.
- **Unrelated:** arm's-length transaction with no observed prior or future reciprocal relationship.

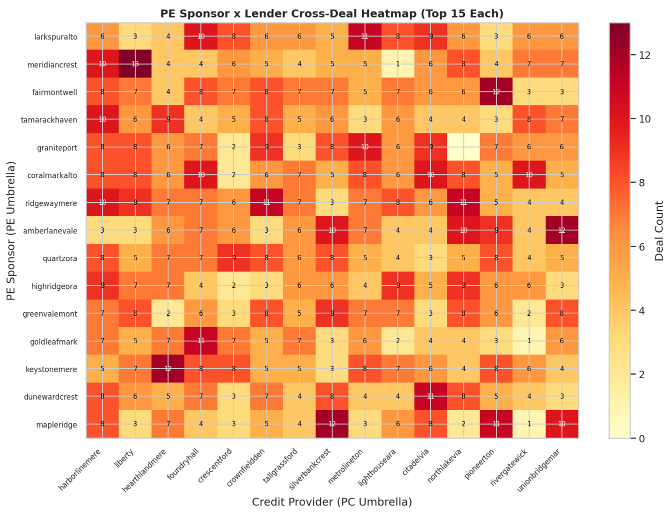

Figure 2: PE Sponsor × Lender Cross-Deal Heatmap (Top 15 Each)

3. Loan-Pricing Analysis

3.1 Descriptive Statistics

Table 2 shows mean and median loan spreads for each relationship category. Affiliated and prior-reciprocal loans carry materially lower spreads than unrelated loans. The unrelated mean of 573.5 bps is approximately 56 bps above the affiliate mean - roughly 10% of the unrelated baseline. Table 2 shows the spread distribution by relationship type. The

Table 2: Loan Spreads by Relationship Type (basis points)

Relationship Type	N	Mean (bps)	Median (bps)	Std Dev
1_Affiliate	2,433	517.2	519.0	101.0
2_Prior_Reciprocal	6,203	563.8	564.0	102.5
3_Reciprocal_Future	1,774	533.5	530.0	97.3
4_Unrelated	4,204	573.5	572.0	102.6

Source: Atlas deal events linked to PE ownership spells. Deals with missing `LoanSpreadBps` excluded.

affiliate category has the lowest median and tightest interquartile range; the unrelated category has the widest spread distribution.

3.2 Regression Results

Table 3 report OLS estimates of loan spread (in basis points) on relationship-type dummies. Model 1 includes only year fixed effects. Model 2 adds borrower controls (log entry EBITDA and entry leverage). Model 3 further adds lender fixed effects. Heteroskedasticity-robust standard errors (HC1) are reported in parentheses. The unrelated category (4_Unrelated) serves as the reference group. All three relationship categories are associated with economically and

Table 3: OLS: Loan Spread (bps) on Relationship Type

Variable (ref: 4_Unrelated)	Model 1	Model 2	Model 3
1_Affiliate	-55.8^{***}	-55.7^{***}	-42.7^{***}
	(2.5)	(2.5)	(9.2)
2_Prior-reciprocal	-15.5^{***}	-16.0^{***}	-2.8
	(2.0)	(2.1)	(9.1)
3_Reciprocal (future)	-19.8^{***}	-19.5^{***}	-7.0
	(2.9)	(3.0)	(9.4)
log(Entry EBITDA)		+1.72	+1.70
		(0.92)	(0.92)
Entry leverage		+0.06	+0.10
		(0.62)	(0.62)
Year FE	Yes	Yes	Yes
Borrower controls	No	Yes	Yes
Lender FE	No	No	Yes
Observations	14,614	14,025	14,025
R^2	0.106	0.105	0.111

Note: Dependent variable is `LoanSpreadBps`. Standard errors are heteroskedasticity-robust (HC1). $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

statistically significant spread reductions in the baseline specification. The affiliate discount is the largest (≈56 bps). The prior-reciprocal discount of 15.5 bps (Model 1), which is constructed using only information available *before* the current deal, represents the preferred causal estimate of relationship lending. Adding borrower controls (Model 2) leaves the coefficients virtually unchanged, indicating that the pricing gap is not driven by observable differences in firm size or leverage. Including lender fixed effects (Model 3) attenuates the coefficients substantially, especially for prior-reciprocal and future-reciprocal relationships, suggesting that part of the raw discount reflects persistent lender-specific pricing styles. Year fixed effects absorb aggregate credit-market conditions. The modest increase in R^2 when moving from Model 1 to Model 3 confirms that lender identity explains additional variation in pricing.

4. Ex-Ante Characteristics and Ex-Post Outcomes

Table 4 compares entry characteristics and exit outcomes across relationship types using the matched Atlas - Summit dataset. Three findings stand out. First, entry characteristics are nearly identical across relationship types - revenue,

Table 4: Entry Characteristics and Exit Outcomes by Relationship Type

Metric	Affiliate	Prior-Recip.	Recip. Future	Unrelated
Entry Revenue ($M)	405.38	398.22	392.67	395.57
Entry EBITDA ($M)	83.48	82.46	82.21	82.20
Gross IRR (%)	0.13	0.13	0.13	0.14
Gross TVM (x)	1.53	1.54	1.53	1.67
Default Rate (%)	5.66	5.50	5.00	5.14
Loss Multiple (x)	0.01	0.01	0.01	0.01

Note: Monetary figures in USD millions. Gross IRR and TVM are deal-level averages from Summit.

EBITDA, and leverage (Figure 3) are statistically indistinguishable, ruling out a simple selection story in which affiliated lenders finance systematically different companies. Second, unrelated deals show the highest gross TVM (1.67x vs. 1.53 - 1.54x), which could reflect that PE sponsors use unrelated lenders in higher-upside situations and accept the higher spread as a trade-off. Third, default rates are modestly lower for reciprocal-future deals (5.0%) relative to affiliates (5.66%), though differences are small across all groups.

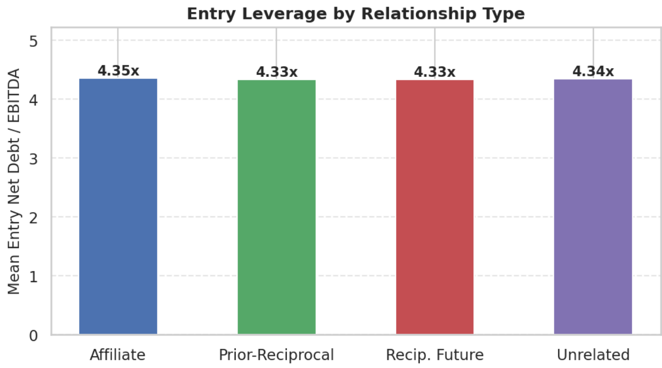

Figure 3: Entry Leverage by Relationship Type

5. Matching Procedure and Empirical Choices

5.1 Atlas - Summit Company Matching

Company names were standardised by lowercasing, removing punctuation, and stripping common legal suffixes. Exact matches on cleaned names formed the high-confidence base. For remaining unmatched Atlas companies, `rapidfuzz` token_sort_ratio with a threshold of 85 generated candidate pairs. Candidates were auto-accepted if the longest meaningful token matched exactly across both names, auto-rejected if it differed, and flagged for manual review otherwise. The match-review file documents all candidate pairs with similarity scores and resolution decisions. Five to seven percent of Summit rows are intentional non-matches. The conservative strategy of treating PENDING rows as rejected unless manually confirmed ensures these non-matches do not contaminate the outcome analysis.

5.2 Key Empirical Choices and Limitations

- **Umbrella definition** is rule-based (first two meaningful tokens after cleaning). While effective for most cases, it is imperfect for complex conglomerate structures.
- **Reciprocity** is measured at the umbrella level rather than the individual fund level. A fund-level definition would be more granular but is not feasible with the available data.
- **Regression controls:** Models 2 and 3 include log(entry EBITDA) and entry leverage. However, other important borrower characteristics (e.g., sector, credit rating, or EBITDA margin) are not available in the Atlas dataset and could further explain the remaining relationship discounts.
- **Outcome comparisons** are based on unadjusted means. Regression-adjusted estimates that control for vintage year, sector, and entry characteristics would provide a cleaner comparison of ex-post performance.

6. Three Additional Empirical Observations

6.1 Relationship Discounts Are Stable Across the Credit Cycle

Computing mean spreads by year and relationship type shows that the affiliate and Prior-reciprocal discounts relative to unrelated deals persist across the 2001 - 2026 sample period, including during the 2008 - 09 financial crisis and the 2020 pandemic shock. Spread levels rise and fall with the credit cycle, but the gaps across relationship types remain approximately constant. This stability suggests the discount is a structural feature of relationship pricing rather than a cyclical phenomenon.

6.2 Entry Leverage Is Uniform Across Relationship Types

Mean entry net debt-to-EBITDA ratios are 4.33 - 4.35 across all four relationship types (Figure 3). The near-identical leverage levels rule out risk-based explanations for the spread differences: affiliated and reciprocal borrowers are no more or less indebted at entry. This result corroborates the inference from Table 4 that pricing differences reflect relationship rents rather than different underlying credit risk.

6.3 PE - Lender Relationships Are Highly Sticky

Figure 4 shows the distribution of time between consecutive deals within PE - lender pairs. The median inter-deal gap is 795 days ($\approx$2.2 years), and the CDF confirms that over 50% of pairs transact again within roughly 3 years. Figure 5 shows that the top 20 PE sponsors by reciprocal deal share all exceed 67%, with *lanterngate* leading at 71.1%.

7. Summary

This analysis finds robust evidence of a loan-spread discount for PE-backed companies that borrow from affiliated or relationship lenders. The affiliate discount of approximately 55.8 basis points and the prior-reciprocal discount of $\approx$15.5 bps (the causally preferred estimate) are large relative to the unrelated baseline of 573.5 bps. Entry characteristics are uniform across relationship types, ruling out simple risk-selection explanations. Relationship pairings are highly persistent, with the typical pair transacting again within about two years. These patterns are consistent with a market in which repeated relationships create bilateral information advantages and relationship rents that PE sponsors and lenders share through favourable pricing.

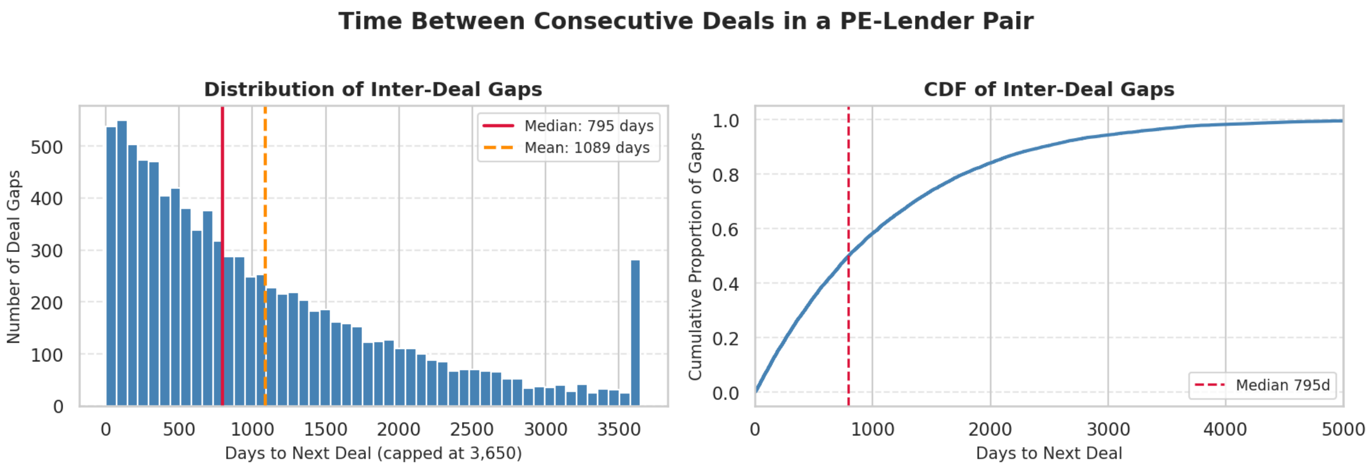

Figure 4: Time Between Consecutive Deals in a PE - Lender Pair

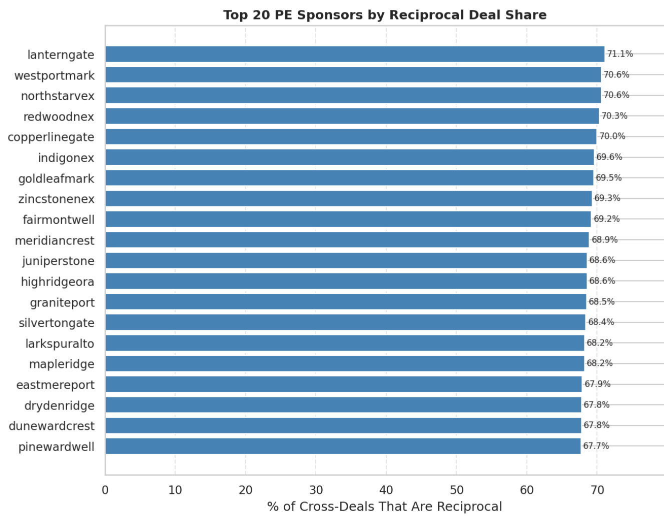

Figure 5: Top 20 PE Sponsors by Reciprocal Deal Share